

A reduced-order integral formulation to account for the finite size effect of isotropic square panels using the transfer matrix method

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Abstract

The transfer matrix method is a well-established prediction tool for the simulation of sound transmission loss and the sound absorption coefficient of flat multilayer systems. Much research has been dedicated to enhancing the accuracy of the method by introducing a finite size effect of the structure to be simulated. The aim of this paper is to present a reduced-order integral formulation to predict radiation efficiency and radiation impedance for a panel with equal lateral dimensions. The results are presented and discussed for different materials in terms of radiation efficiency, sound transmission loss and the sound absorption coefficient. Finally, the application of the proposed methodology for rectangular multilayer systems is also investigated and validated against experimental data.

Keywords: radiation efficiency, radiation impedance, finite size correction, transfer matrix

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I. INTRODUCTION

Multilayer panels are largely used for noise and vibration control with several fields of application including automotive, building and architectural acoustics. The prediction of the performance of these systems has been widely investigated in the literature using different numerical approaches such as the finite element method, the statistical energy analysis, and the transfer matrix method, etc. In particular, when dealing with flat structures, the transfer matrix method has proved to be a fast and accurate technique for identifying the main acoustical indicators as radiation efficiency, sound transmission loss and sound absorption coefficient.

In the original formulation¹ of the transfer matrix method, the structure to be simulated is assumed to have an infinite extent. Generally, some discrepancies at mid to low frequencies can be observed in these conditions for sound transmission and absorption applications when the predictions are compared with experimental tests. With this in mind, several works¹⁻⁵ have focused on introducing corrections to account for the geometrical finite size effect for an acoustical excitation. In particular Villot *et al.*² proposed an expression for radiation impedance derived in the wave number domain and Vigran³ suggested a one-dimensional simplification of formulation proposed in ref. [2] for isotropic materials having comparable lateral dimensions. Ghinet and Atalla⁴ proposed a more general expression for radiation impedance presented in ref. [1], while Rhazi and Atalla⁵ presented an analytical simplification of the expression in proposed in ref. [4] allowing for fast computation of the radiated and transmitted power of multi-layered panels. A review and the relevance of such formulations will be given below. However, a common limitation of such formulations is the high computational time required even for simple structures. Rhazi and Atalla⁶ recently proposed the same concept of correction to model the vibroacoustic response of multi-material structures under mechanical excitation. In this paper, a revised formulation for the calculation of the radiation efficiency and radiation impedance of a square multilayer panel will be presented and compared with experimental data and existing techniques.

The paper is organized as follows: the description of the methods proposed in the literature together with the revised formulation to determine the sound absorption coefficient and transmission loss will be presented in the next section. Section III will compare the accuracy and computational time of the proposed formulation and existing methods in sound transmission and absorption applications. The application of the proposed methodology to rectangular shaped structures will be discussed in

Section IV, and validation against experimental tests will also be presented. Finally, concluding remarks will be made in the last section.

II. MATERIALS AND METHODS

A. THE TRANSFER MATRIX METHOD

The transfer matrix method¹ (hereafter TMM) has been developed for the study of plane wave propagation within multilayer systems (made of elastic solids, rigid and elastic framed porous materials, membranes, resonators, porous and impervious sheets, etc.).

Following the transfer matrix calculation scheme, considering a multilayer of an infinite extent, it is possible to calculate the sound absorption coefficient for a plane wave impinging on it at incident angle θ as follows:

$$\alpha(\theta) = 1 - |R(\theta)|^2 = 1 - \left| \frac{Z_s(\theta)\cos(\theta) - Z_0}{Z_s(\theta)\cos(\theta) + Z_0} \right|^2 \quad (1)$$

R [-] being the complex reflection coefficient, Z_0 [rayls] the characteristic impedance for the air and Z_s [rayls] the surface acoustic impedance of the medium given by the ratio between the sound pressure and particle velocity at the first layer of the panel.

Using the same scheme, the transmission loss of a system for a plane wave impinging on it at an incident angle θ is calculated as follows:

$$TL(\theta) = 10 \log_{10} \left(\frac{1}{|T(\theta)|^2} \right) \quad (2)$$

T [-] being the complex transmission coefficient determined directly by solving a global matrix depending on physical and mechanical properties of each material constituting the multilayer panel.

The transfer matrix method is quite general and can be used to simulate normal and oblique incidence as well as diffuse field sound absorption and transmission loss. It is worth mentioning that this approach deals with structures with infinite lateral size. Geometrical corrections have been widely investigated in the literature¹⁻⁵ to account for the finite size and all corrections are based on the concepts of (i) the radiation efficiency $\sigma_R(\theta, \varphi)$ of an equivalent baffled window for sound transmission loss problems and (ii) the radiation impedance $Z_R(\theta, \varphi)$ for the sound absorption coefficient, which are recalled in Section II.B. The radiation efficiency is a commonly

used quantity to characterize the efficiency of a given vibrating surface, as a sound radiator and by definition:

$$\sigma_R = \frac{\Pi_{rad}}{Z_0 S \langle V^2 \rangle} = \frac{\Re(Z_R)}{Z_0} \quad (3)$$

where Π_{rad} [W] is the radiated power from the actual vibrating surface, having the area S [m²], to the surrounding medium with characteristic impedance Z_0 . The quantity $\langle V^2 \rangle$ is the mean square velocity amplitude taken over the surface. The denominator in the expression is the power radiated from a partial area S of an infinitely large plane surface, all parts vibrating in phase with a velocity equal to this mean value (that is a plane wave radiation condition).

For vibrating structures, the radiation impedance is defined as the ratio of the average acoustic pressure to the volume velocity,

$$Z_R = \int_0^a \int_0^b \int_0^a \int_0^b W_n(x, y) G(x, y, x', y') W_n^*(x', y') dx dy dx' dy' \quad (4)$$

where, a and b are the length and the width of the structure, respectively, W_n is the plate displacement field and G the Green function expressed as:

$$W_n(x, y) = e^{-j(k_t x \cos(\varphi) + k_t y \sin(\varphi))}$$

$$G(x, y, x', y') = \frac{e^{-jk_0 \sqrt{(x-x')^2 + (y-y')^2}}}{2\pi \sqrt{(x-x')^2 + (y-y')^2}} \quad (5)$$

k_0 [m⁻¹] the wave number in free air and $k_t = k_0 \cdot \sin \theta$ [m⁻¹] the structural wave number.

For practical applications, in order to reduce computational time, radiation efficiency and radiation impedance are averaged in the φ domain between 0 and 2π , obtaining two functions $\sigma_{R,avg}(\theta)$ and $Z_{R,avg}(\theta)$ of the heading angle θ . In this paper, all comparisons will be presented for averaged radiation efficiency and radiation impedance thus limiting analyses to isotropic structures.

When introducing the finite size corrections, the diffuse field incidence sound absorption coefficient and sound transmission loss are calculated as follows:

$$\alpha_d = \frac{\int_0^{\theta_{lim}} \frac{\Re\{\zeta_s(\theta)\}}{|\zeta_s(\theta) + \zeta_{R,avg}(\theta)|^2} \sin \theta d\theta}{\int_0^{\theta_{lim}} \cos \theta \sin \theta d\theta} \quad (6)$$

$$TL_d = -10 \log_{10} \left[\frac{\int_0^{\theta_{lim}} |T(\theta)|^2 \sigma_{R,avg}(\theta) \cos^2 \theta \sin \theta d\theta}{\int_0^{\theta_{lim}} \cos \theta \sin \theta d\theta} \right] \quad (7)$$

with $\zeta_s = Z_s/Z_0$ and $\zeta_{R,avg} = Z_{R,avg}/Z_0$ and θ_{lim} being equal to 90° for sound absorption and 78° for sound transmission loss, as suggested in ref. [1]. Several analyses have underlined that a better match between experimental tests and numerical simulations can be achieved when such corrections are applied. As an alternative formulation of Eq. (7), Villot *et al.*² suggest the use of the same size correction for the incident power. However, as stated in ref. 1, contrary to the input power, the incident power is independent of edge effects. Thus in the present research the finite size correction has been applied only to the transmitted power.

Throughout the paper, FTMM will be used to indicate the application of the finite size correction to the transfer matrix simulation.

B. GEOMETRICAL CORRECTIONS TO ACCOUNT FOR THE FINITE SIZE EFFECT.

A review of existing finite size correction will be presented in this section for a comparison with the proposed scheme described in Section II. C.

For sound transmission loss problems, Villot *et al.*² proposed an expression for radiation impedance derived in the wave number domain. The averaged correction factor is calculated as follows:

$$\sigma_{R,avg}(k_t) = \frac{1}{2\pi} \frac{S}{\pi^2} \int_0^{k_0} \int_0^{2\pi} \int_0^{2\pi} \frac{1 - (\cos(k_r \cos \psi - k_t \cos \varphi) L_x)}{[(k_r \cos \psi - k_t \cos \varphi) L_x]^2} \times \frac{1 - (\cos(k_r \sin \psi - k_t \sin \varphi) L_y)}{[(k_r \sin \psi - k_t \sin \varphi) L_y]^2} \times \frac{k_0 k_r}{\sqrt{(k_0^2 - k_r^2)}} d\varphi d\psi dk_r \quad (8)$$

L_x [m] and L_y [m] being the lateral dimension of the structure, $S = L_x \cdot L_y$ and k_r represents the modulus of the wavenumber spectrum of the velocity field of the structure.

Vigran³ proposed a one-dimensional simplification of Eq. (8) for isotropic materials having comparable lateral dimensions:

$$\sigma_{R,avg}(k_t) = \frac{L k_0}{2\pi} \int_0^{k_0} \frac{\sin^2 \left[(k_r - k_t) \frac{L}{2} \right]}{\left[(k_r - k_t) \frac{L}{2} \right]^2 \cdot \sqrt{k_0^2 - k_r^2}} dk_r \quad (9)$$

L being equal to $\sqrt{L_x \cdot L_y}$.

Ghinet and Atalla⁴ proposed a more general expression for radiation impedance which can be expressed as follows in a condensed form:

$$Z_{R,avg}(k_t) = j\rho_0\omega \frac{L_y}{8\pi^2} \int_0^2 \int_0^2 \int_0^{2\pi} \frac{e^{jA\sqrt{u^2+Bu'^2}}}{\sqrt{u^2+Bu'^2}} (2-u)(2-u') \times \cos(Cu \cos \varphi) \cos(Du' \sin \varphi) d\varphi du' du \quad (10)$$

ρ_0 [kg/m³] being the density of air and $r=L_x/L_y$, $A=-k_0 \cdot L_x/2$, $B=(L_y/L_x)^2$, $C=k_t \cdot L_x/2$ and $D=k_t \cdot L_x/(2r)$.

According to Eqs. (6) and (7), Eq. (10) can be utilized both in sound absorption problems as well as in sound transmission problems recalling that σ_R equals the real part of Z_R/Z_0 .

Apart from Eq. (9), a common limitation of these methods is their high computational cost when calculated in the θ domain for each frequency of interest. Having this in mind, Rhazi and Atalla⁵ presented an analytical simplification of the expression in Eq. (10) allowing for fast computation of the radiated and transmitted power of multi-layered panels. In particular, the authors were able to calculate the integral Eq. (10) analytically using two changes of variables from (u, u') to $(X=u, Y=u' \cdot L_y/L_x)$ and from (X, Y) to $(R \cos \psi, R \sin \psi)$ and they proposed the following expression for radiation impedance:

$$Z_{R,avg}(k_t) = j\rho_0\omega \frac{L_y}{8\pi^2} \int_0^{\pi/2} \int_0^{2\pi} r F_R(\psi) + jr F_I(\psi) d\psi d\varphi \quad (11)$$

with F_R and F_I being functions of the variable ψ solved analytically⁵.

C. PROPOSED METHODOLOGY

In this paper, a further simplification of Eq. (10) will be proposed. In fact, making the change in variables:

$$\begin{cases} u = \frac{k}{\sqrt{B \tan^2 \xi + 1}} \\ u' = \frac{k \tan \xi}{\sqrt{B \tan^2 \xi + 1}} \end{cases} \quad (12)$$

it is possible to prove that:

$$Z_{R,avg}(k_t) = j\rho_0\omega \frac{L_y}{8\pi^2} \left(\int_0^2 \int_0^{\pi/2} \int_0^{2\pi} F(k, \xi, \varphi) dk d\xi d\varphi + \int_2^{2\sqrt{B}} \int_{\xi_1}^{\pi/2} \int_0^{2\pi} F(k, \xi, \varphi) dk d\xi d\varphi + \int_{2\sqrt{B}}^{2\sqrt{B+1}} \int_{\xi_1}^{\xi_2} \int_0^{2\pi} F(k, \xi, \varphi) dk d\xi d\varphi \right), \quad (13)$$

where:

$$\begin{aligned}
 \xi_1 &= \tan^{-1} \left(\sqrt{\frac{k^2 - 4}{4B}} \right) \\
 \xi_2 &= \tan^{-1} \left(\frac{2}{\sqrt{k^2 - 4}} \right) \\
 F(k, \xi, \varphi) &= \frac{e^{jAk}}{B - (B-1)\cos^2 \xi} \left(2 - \frac{k}{\sqrt{B \tan^2 \xi + 1}} \right) \left(2 - \frac{k \tan \xi}{\sqrt{B \tan^2 \xi + 1}} \right) \\
 &\quad \times \cos \left(\frac{Ck \cos \varphi}{\sqrt{B \tan^2 \xi + 1}} \right) \cos \left(\frac{Dk \tan \xi \sin \varphi}{\sqrt{B \tan^2 \xi + 1}} \right).
 \end{aligned} \tag{14}$$

In this form Eq. (13) is able to provide faster computation if compared with Eq. (10). In fact, the change in variables in Eq. (12) allows for the removal of singularity in $u=u'=0$ of Eq. (10). Furthermore, the first term in the integral of Eq. (10) is characterized by large oscillations in an elliptic region ($u^2 + Bu'^2 = k^2$) when A (i.e. frequency) increases; the proposed change in variables makes maxima and minima of such oscillation varying in a linear region parallel to the ξ axis and allowing for a faster computation of the integral in Eq. (13). An example of computation efficiency of Eqs. (13) and (10) will be given in Section III.

For the special case of $L_x=L_y$, $B=1$ and $C=D$ and Eq. (13) reduces to:

$$\begin{aligned}
 Z_{R,avg}(k_t) &= j\rho_0\omega \frac{L_y}{8\pi^2} \left(\int_0^2 \int_0^{\pi/2} \int_0^{2\pi} F'(k, \xi, \varphi) dk d\xi d\varphi + \right. \\
 &\quad \left. + \int_2^{2\sqrt{2}} \int_{\xi_1}^{\xi_2} \int_0^{2\pi} F'(k, \xi, \varphi) dk d\xi d\varphi \right).
 \end{aligned} \tag{15}$$

where:

$$\begin{aligned}
 F'(k, \xi, \varphi) &= e^{jAk} (2 - k \cos \xi)(2 - k \sin \xi) \\
 &\quad \times \cos(Ck \cos \xi \cos \varphi) \cos(Ck \sin \xi \sin \varphi).
 \end{aligned} \tag{16}$$

Firstly let us consider integration in the φ domain; using the relation $\cos(a) \cdot \cos(b) = 1/2[\cos(a+b) + \cos(a-b)]$ it is possible to demonstrate that:

$$\int_0^{2\pi} \cos(\alpha' \cos \varphi) \cdot \cos(\beta' \sin \varphi) d\varphi = 2\pi \cdot \Re \{ I_0(jCk) \}, \tag{17}$$

I_0 being the modified Bessel function of the first kind, $\alpha' = C \cdot k \cdot \cos \xi$ and $\beta' = C \cdot k \cdot \sin \xi$.

Eq. (17) suggests that integration in the φ domain can be solved analytically and the results only depend on variable k . In addition, integration in the ξ domain admits exact analytical solutions for both integrals in Eq. (15). As a conclusion for the

special case of square flat panels, the triple integral in Eq. (15) can be reduced to the single integral in the k domain which can be solved numerically:

$$Z_{R,avg}(k_t) = j\rho_0\omega \frac{L_y}{4\pi} \left(\int_0^2 e^{jAk} \left(4k - \frac{(k^2 \cos^2 \xi)}{2} + 2k \cos \xi - 2k \sin \xi \right) \right)_{\xi_1}^{\pi/2} \cdot \Re \{ I_0(jCk) \} dk + \int_2^{2\sqrt{2}} e^{jAk} \left(4k - \frac{(k^2 \cos^2 \xi)}{2} + 2k \cos \xi - 2k \sin \xi \right) \right)_{\xi_1}^{\xi_2} \cdot \Re \{ I_0(jCk) \} dk \right). \quad (18)$$

The accuracy of the proposed methodology for panels with different lateral dimensions will also be investigated in section IV.

III. RESULTS

Some comparisons between the afore-mentioned formulations will be presented in this section and then discussed in terms of the radiation efficiency of a given structure for bending waves, diffuse field sound transmission loss and the sound absorption coefficient. All computational time calculations refer to a 2GHz PC Intel Core I7 -6 GB RAM. The multiple integrals in Eqs. (9)–(11), (13) and (18) have been solved numerically using a global adaptive quadrature approach (implemented in *@integral*, *@integral2* and *@integral3* functions of Matlab[®]), setting absolute and relative tolerance to $1e^{-3}$.

A. RADIATION EFFICIENCY FOR BENDING WAVES

It is well known¹ that by replacing the trace wave number k_t in Eqs. (8)–(11) and (18) with the bending wave number $\left(\sqrt{\omega^4 m/D} \right)$, D being the bending stiffness and m the surface density, the same equations lead to the radiation efficiency of the structure for such waves. Figure 1 depicts the comparison between different finite size effect formulations in terms of radiation efficiency for an aluminum square plate of $1 \times 1 \text{ m}^2$. The mechanical and geometrical properties of the plate are summarized in Table 1. Simulations have been carried out in 1/3 octave frequency bands between 100 Hz and 10 kHz considering 9 frequencies per 1/3 octave band.

From the data in Figure 1, it is possible to observe significant deviations at low frequencies ($< 500 \text{ Hz}$) when the Vigran formulation (Eq. (9)) is used. Moreover, the formulation proposed by Rhazi-Atalla (Eq. (11)) is characterized by a deviation of around 0.5 dB at frequencies lower than 1 kHz when compared with the Ghinet-Atalla method (Eq. (10)). Finally, the proposed methodology (Eq. (18)) leads to

identical results of the Ghinet and Atalla method (Eq. (10)) and Villot approach (Eq. (8)). Here it is worth mentioning that Rhazi-Atalla and the proposed formulation lead to the same results when Eq. (18) is solved in the k domain between 0 and 2 only. Regarding computational times, the proposed methodology and Vigran approach offer faster results (between 0.1 and 0.2 s), the Rhazi-Atalla method is able to give results in around 50 s, while the Villot approach and Ghinet-Atalla formulations require considerable computational time (> 10 minutes).

B. SOUND TRANSMISSION LOSS

The same aluminum plate from the previous subsection was used to compare different formulations in terms of sound transmission loss. Simulations were carried out in 1/3 octave frequency bands between 50 Hz and 10 kHz, considering 9 frequencies per 1/3 octave band, and integrating a vector of evenly spaced 60 angles between 0 and 78°. The curves of sound transmission loss are shown in Figure 2. The sound transmission loss of the infinite extent plate (indicated as TMM) has been calculated using expression Eq. (7) omitting the term $\sigma_R \cos(\theta)$.

Once again, the accuracy of the proposed methodology was comparable with the Villot (Eq. 5) and Rhazi-Atalla (Eq. 8) formulations (here used for determining the radiation efficiency) but the computational time was highly reduced. The Vigran approach seems to be the fastest method to account for the finite size correction in sound transmission problems; however, for the analyzed example, it was characterized by deviations between 0.5 and 3 dB at frequencies lower than 100 Hz.

C. SOUND ABSORPTION COEFFICIENT

In order to investigate the accuracy of the proposed methodology in sound absorption problems the next examples consider the prediction of the diffuse field sound absorption coefficient measured in a reverberant chamber according to ASTM C423⁷ or ISO 354⁸. In particular, Figure 3 compares experimental data on a porous foam carried out in a small reverberant room at the Université de Sherbrooke¹ with formulations in Eqs. (11) and (18). The geometrical and physical data of the foam are summarized in Table 1 and the foam was simulated using Johnson-Champoux-Allard model with a limp frame approximation (derived from the Biot theory assuming that the stiffness of the frame is negligible)¹. The size of the foam was 1.829 x 1.829 m². Simulations were carried out in 1/3 octave frequency bands between 100 Hz and 5

kHz, considering 9 frequencies per 1/3 octave band, and integrating a vector of evenly spaced 60 angles between 0 and 90°.

The data in Figure 3 show comparable accuracy between both the analytical formulations and a satisfying agreement with experimental data. In particular the average absolute difference between sound absorption coefficients calculated using analytic formulations is equal to 0.002, while the same quantity related to the proposed formulation and the experimental data is equal to 0.07. However, the computational time is considerably reduced when the proposed methodology is applied. The sound absorption coefficient of the infinite extent plate (indicated as TMM) has been calculated using expression Eq. (6) considering $Z_R = 1/\cos(\theta)$ ¹.

The second example refers to the comparison between experimental data on porous samples made of polyester fiber (25 mm thick) carried out in a reverberant chamber with a volume of 250 m³ and the formulations in Eqs. (11) and (18). The physical and geometrical proprieties of polyester fiber are summarized in Table 1 and polyester fiber was simulated as a rigid frame porous material using a semi-empirical model⁹ to determine its characteristic impedance and complex wave number.

The size of the foam was 3.5 x 3.5 m² and the results in terms of accuracy and computational time were similar to the one investigated in the previous example. Here it is worth mentioning that the computational times reduced from 12 hours to 3 hours when the diffuse field sound absorption coefficient of the fiber material was simulated using the formulations in Eqs. (10) and (13), respectively.

IV. APPLICATION OF THE PROPOSED METHODOLOGY TO RECTANGULAR MULTILAYER PANELS

As stated in Section III.C, the proposed simplified formulation (Eq. (18)) to account for the finite size effect in sound absorption and transmission problems is strictly valid for square systems. In this section, the possibility of applying the formulation to panels with different lateral sizes has been investigated, replacing L_x and L_y with $L = \sqrt{L_x \cdot L_y}$. This strategy is the same as that proposed in Eq. (9).

The application of Eq. (18) with $L_x = L_y = L$ surely leads to values of real and imaginary part of radiation impedance which differ in $\Delta\Re$ and $\Delta\Im$ if compared with values obtained using Eqs. (10) or (13). Thus for a given frequency f and heading angle θ it is possible to write:

$$\begin{aligned}\zeta_{R,avg}(f, \theta) \Big|_{Eq.(18)} &= \zeta_{R,avg}(f, \theta) \Big|_{Eq.(13)} + (\Delta \Re(f, \theta) + j\Delta \Im(f, \theta)) \\ &= \zeta_{R,avg}(f, \theta) \Big|_{Eq.(13)} + \Delta Z(f, \theta)\end{aligned}\quad (19)$$

The subscripts in Eq. (19) indicate the equations used for determining the radiation impedance.

Starting from Eq. (6), an estimation of the error can be obtained by examining the integrand function; in particular, the aim of the proposed analysis is to find for which aspect ratio, frequency and heading angle the following condition is satisfied:

$$\left| \frac{1}{\left| \zeta_s(f, \theta) + \zeta_{R,avg}(f, \theta) \right|^2} - \frac{1}{\left| \zeta_s(f, \theta) + \zeta_{R,avg}(f, \theta) + \Delta Z(f, \theta) \right|^2} \right| \leq \delta \quad (20)$$

δ being an arbitrarily fixed small real number.

If it is chosen:

$$\delta = \frac{1}{\varepsilon} \frac{1}{\left| \zeta_s(f, \theta) + \zeta_{R,avg}(f, \theta) \right|^2} \quad (\varepsilon \geq 1) \quad (21)$$

then Eq. (20) is equivalent to calculate the relative error on the sound absorption coefficient:

$$\frac{\left| \alpha(f, \theta) \Big|_{Eq.(13)} - \alpha(f, \theta) \Big|_{Eq.(18)} \right|}{\alpha(f, \theta) \Big|_{Eq.(13)}} \leq \frac{1}{\varepsilon} \quad (22)$$

In its current form, Eq. (22) suggests that in order to estimate the error when the simplification described for rectangular structures is used, it is necessary to know the surface impedance of the multilayer system in advance.

Omitting (f, θ) , after simple mathematical manipulations, Eq. (20) can be written as:

$$\left| \zeta_s + \zeta_{R,avg} \Big|_{Eq.(13)} + \Delta Z \right|^2 \leq \lambda_1^2 \left| \zeta_s + \zeta_{R,avg} \Big|_{Eq.(13)} \right|^2 \quad (23)$$

$$\left| \zeta_s + \zeta_{R,avg} \Big|_{Eq.(13)} + \Delta Z \right|^2 \geq \lambda_2^2 \left| \zeta_s + \zeta_{R,avg} \Big|_{Eq.(13)} \right|^2 \quad (24)$$

where $\lambda_1 = \sqrt{\varepsilon/(\varepsilon-1)}$ and $\lambda_2 = \sqrt{\varepsilon/(\varepsilon+1)}$.

Equations (23) and (24) indicate that in a complex Gauss plane all points having coordinates $(\Delta \Re; \Delta \Im)$ lying within the annulus with the center in

$$\left(\Re \left(\zeta_s + \zeta_{R,avg} \Big|_{Eq.(13)} \right); \Im \left(\zeta_s + \zeta_{R,avg} \Big|_{Eq.(13)} \right) \right) \quad (25)$$

and radii equal to

$$R_1 = \lambda_1 \left| \zeta_s + \zeta_{R,avg} \Big|_{Eq.(13)} \right| \text{ and } R_2 = \lambda_2 \left| \zeta_s + \zeta_{R,avg} \Big|_{Eq.(13)} \right| \quad (26)$$

will satisfy Eq. (22). A graphical representation of the complex Gauss plane is depicted in Figure 5a.

Since all terms in the first equation of Eq. (23) are positive, such equation is satisfied if:

$$|\zeta_s + \zeta_{R,avg} + \Delta Z| \leq \lambda_1 |\zeta_s + \zeta_{R,avg}| \quad (27)$$

and by applying triangular inequalities in the complex domain (recalling that the real parts of ζ_s and $\zeta_{R,avg}$ are positive) it can be proved that a subset of error coordinates $(\Delta \Re; \Delta \Im)$ which surely will satisfy Eq. (22) is given by:

$$\frac{|\Delta Z|}{\left| \Re(\zeta_{R,avg}|_{Eq.(13)}) \right|} = \frac{|\zeta_{R,avg}|_{Eq.(13)} - \zeta_{R,avg}|_{Eq.(18)}|}{\left| \Re(\zeta_{R,avg}|_{Eq.(13)}) \right|} \leq (\lambda_1 - 1) \quad (28)$$

Using similar considerations for Eq. (23), it can be verified that it is satisfied if $\lambda_2 \leq 1$, which is always true since $\varepsilon \geq 1$. Eq. (28) indicates that in a complex Gauss plane, the solutions lie within a circumference having the center at (0;0) and radius equal to

$$\sqrt{(\lambda_1 - 1) \cdot \left| \zeta_s + \zeta_{R,avg}|_{Eq.(13)} \right|} \quad (\text{Figure 5b}).$$

Although Eq. (28) represents a possible overestimation of the exact error ΔZ , such expression is no longer dependent on the surface impedance of the multilayer systems. In particular, the ratio in Eq. (28) can be calculated numerically by solving Eq. (13) with $L_x \neq L_y$ and Eq. (18) with $L = \sqrt{L_x \cdot L_y}$ for any frequency f and heading angle θ of interest. As an example, Figure 6 shows the values of relative error $1/\varepsilon$ (in percentage) as a function of heading angles for frequencies equal to 100 Hz, 500 Hz, 1000 Hz and 5000 Hz for two different ratios between sides. The panels have dimensions of $L_x=4$ m and $L_y=2.7$ m (thus a ratio of 1.5 between sides) and $L_x=4$ m and $L_y=2$ m (thus a ratio of 2 between sides). The last case can be considered as a limited case if compared with the sizes suggested in ISO 354 Standard.

From the data depicted in Figure 6, it is possible to observe that the values of relative errors are lower than 20 % for each investigated frequency and heading angle. Generally when the frequency increases the relative error decreases.

Using the same approach, it is possible to obtain an estimation of error for the case of sound transmission loss solving the following expression:

$$\frac{\left| \Re(\zeta_{R,avg}(f, \theta)|_{Eq.(13)}) - \Re(\zeta_{R,avg}(f, \theta)|_{Eq.(18)}) \right|}{\left| \Re(\zeta_{R,avg}(f, \theta)|_{Eq.(13)}) \right|} \leq \frac{1}{\varepsilon} \quad (29)$$

As an example, Figure 7 shows the values of relative error $1/\varepsilon$ (in percentage) as a function of heading angles for frequencies equal to 100 Hz, 500 Hz, 1000 Hz and 5000 Hz. The panel has dimensions of $L_x=2$ m and $L_y=1$ m (thus a ratio of 2 between

sides). It is worth mentioning that such ratio is generally higher than the one used in practical applications in the automotive field.

From the data depicted in Figure 7, it is possible to observe that values of relative errors are lower than 10 % for each investigated frequency and heading angle. Similarly, to the sound absorption coefficient when the frequency increases the relative error decreases.

Figures 8 and 9 compare the experimental test and analytical prediction in terms of diffuse field sound transmission loss for a 1.8 mm thick steel plate (Table 1) and the diffuse field sound absorption coefficient for a 90 mm thick rock wool material (Rockwool® 225 in Table 1). Eqs. (11) (for the case $L_x \neq L_y$) and (18) (using $L = \sqrt{L_x \cdot L_y}$) have been utilized for the comparisons.

In particular, the experimental test on the steel plate has been carried out on a rectangular panel (lateral size of $1.21 \times 0.95 \text{ m}^2$) according to ISO 15186-1:2003 Standard¹⁰ using sound intensity measurements. The experimental tests on mineral wool were performed in a reverberant chamber with a volume of 210 m^3 on a sample with lateral dimensions of $3.6 \times 3 \text{ m}^2$ following Mounting A of ISO 354 Standard. The rock wool material was simulated as a rigid frame porous medium with complex properties depending on airflow resistivity¹¹; moreover, for a better match with experimental data the size effect was also applied to incident power as suggested in ref. [1]. In both cases, it is possible to observe a good correlation between the experimental data and prediction and a satisfying accuracy of the proposed methodology in the case of rectangular systems when compared to Eq. (11).

Finally, Figure 10 shows the comparison between the experimental test and analytical prediction in terms of diffuse field sound absorption coefficient for a multilayer system having size of $4 \times 3 \text{ m}^2$ and made of a porous material (density of 190 kg/m^3 and 15 mm thick) and an air gap having a thickness of 18 mm. The porous material was simulated using Johnson-Champoux-Allard model; geometrical and physical data of the material are summarized in Table 1 and have been determined using an inverse procedure starting from minimisation of complex acoustical parameters (characteristic impedance, complex wave number, surface impedance) as described in ref. [12]. Once again it is possible to observe a satisfying accuracy of the proposed methodology in prediction of the diffuse field sound absorption coefficient.

V. CONCLUSIONS

A reduced order integral formulation to calculate average radiation impedance and radiation efficiency has been proposed in this paper to account for the finite size effect of flat multilayer panels using the well-established transfer matrix method. The formulation is strictly valid for squared panels and it has been derived from a more efficient revision of the exact expression of the radiation impedance of rectangular panels. From comparison with different formulations proposed in the literature, the proposed methodology has demonstrated high accuracy and a much lower computational time. In fact, the proposed formulation has been proved to allow for the prediction of sound transmission loss and sound absorption coefficient in a time reduced of at least two order of magnitude when compared with the faster methodology proposed in literature at the present.

The proposed methodology has also been applied to rectangular systems and unless the lateral dimensions are greatly different, the uncertainties in terms of the sound absorption coefficient and transmission loss are negligible for practical applications.

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Table 1. Parameters of the plates and porous materials used for comparisons.

	d	ρ	E	ν	η	σ	ϕ	α_∞	Λ	Λ'
	[mm]	[kg/m ³]	[Pa]	[-]	[-]	[Pas/m ²]	[-]	[-]	[μ m]	[μ m]
Aluminum Plate	5	2700	69e9	0.3	0.05	-	-	-	-	-
Foam	76.2	8.8	-	-	-	10900	0.99	1	100	130
Polyester fiber	25	30	-	-	-	3100	-	-	-	-
Steel Plate	1.8	7850	2e11	0.33	0.05	-	-	-	-	-
Rockwool 225	90	70	-	-	-	32100	-	-	-	-
Porous material	15	190	-	-	-	108000	0.99	1.1	13	100

Figure 1 - Radiation efficiency for bending waves of a 5 mm thick aluminum plate. Comparison between different finite size effect formulations and computational times.

Figure 2 - Diffuse field sound transmission loss of a 5 mm thick aluminum plate. Comparison between different finite size effect formulations and computational times.

Figure 3 - Diffuse field sound absorption coefficient of a 76.2 mm porous foam. Comparison between different finite size effect formulations and computational times.

Figure 4 - Diffuse field sound absorption coefficient of a 25 mm polyester fiber material. Comparison between different finite size effect formulations and computational times.

Figure 5 - Complex Gauss plane representation of the error ΔZ . (a) Complete set of values satisfying Eq. (22). (b) Subset of values satisfying Eq. (22).

Figure 6 - Values of relative error $1/\varepsilon$ for the sound absorption coefficient as a function of heading angles θ at frequencies equal to 100 Hz, 500 Hz, 1000 Hz and 5000 Hz for panels having dimensions of (a) $L_x=4$ m and $L_y=2.7$ m and (b) $L_x=4$ m and $L_y=2$ m.

Figure 7 - Values of relative error $1/\varepsilon$ for sound transmission coefficient as a function of heading angles θ at frequencies equal to 100 Hz, 500 Hz, 1000 Hz and 5000 Hz for a panel having dimensions of $L_x=2$ m and $L_y=1$ m.

Figure 8 - Diffuse field sound transmission loss of a 1.21×0.95 m² steel plate. Comparison between experimental data and analytical predictions.

Figure 9 - Diffuse field sound absorption coefficient of a 3.6×3 m² Rockwool[®] 225 material. Comparison between experimental data and analytical predictions.

Figure 10 - Diffuse field sound absorption coefficient of a 4×3 m² multilayer system made of porous material and air gap. Comparison between experimental data and analytical predictions.





















